

The Modigliani and Miller Proposition 2

➤ Modigliani and Miller Proposition 2 states that the cost of (required return on) a firm's ordinary shares is directly related to the debt-to-equity ratio.

➤ If a firm has one type of equity:

$$WACC = x_{Debt}k_{Debt} + x_{os}k_{os} \quad (16.2)$$

➤ If we rearrange equation 16.2, we have:

$$k_{os} = k_{Assets} + \left(\frac{V_{Debt}}{V_{os}} \right) (k_{Assets} - k_{Debt}) \quad (16.3)$$

The Modigliani and Miller Propositions

M&M Proposition II Example

After its restructuring, Millennium Motors will be financed with 20 per cent debt and 80 per cent ordinary shares. The return on assets is 10 per cent and the return on debt is 5 per cent. What is the cost of equity for the firm?

$$\begin{aligned}k_{os} &= 0.10 + \left(\frac{0.2}{0.8}\right)(0.10 - 0.05) \\&= 0.1125, \text{ or } 11.25\%\end{aligned}$$

The Modigliani and Miller Propositions

- The first source of risk is business risk. It is the risk associated with the characteristics of the firm's business activities.
- The second source of risk is the capital structure of the firm which reflects the effect that the firm's financing decisions have on the riskiness of the cash flows that the shareholders will receive.
- Financial risk is associated with required payments to a firm's lenders.

Exh 16.2: Relations between Business Risk, Financial Risk and Total Equity Risk

EXHIBIT 16.2

RELATIONS BETWEEN BUSINESS RISK, FINANCIAL RISK AND TOTAL EQUITY RISK

Revenue	}	Business risk
– Cost of goods sold		
Gross profit	}	x
– Selling, general and admin. expenses		
Operating profit	}	Financial risk
– Interest expense		
Earnings before tax	}	=
– Income tax		
Net income	}	Total equity risk

The total risk associated with the cash flows that shareholders are entitled to receive reflects the risk related to the firm's assets (business risk) and the risk related to the way those assets are financed (financial risk). (We assume here that net income (net profit or net earnings) is a reasonable measure of these cash flows.)

Exh I 6.3: Illustration of Relations between Business Risk, Financial Risk and Total Risk

EXHIBIT 16.3

ILLUSTRATION OF RELATIONS BETWEEN BUSINESS RISK, FINANCIAL RISK AND TOTAL RISK

	No Financial Risk				Financial Risk			
	Low Operating Leverage		High Operating Leverage		Low Operating Leverage		High Operating Leverage	
Column	1	2	3	4	5	6	7	8
Variable costs as % of total costs	80%		40%		80%		40%	
Interest expense	€0.00		€0.00		€15.00		€15.00	
	Before	After	Before	After	Before	After	Before	After
Revenue	€100.00	€80.00	€100.00	€80.00	€100.00	€80.00	€100.00	€80.00
– Cost of goods sold (VC)	60.00	48.00	30.00	24.00	60.00	48.00	30.00	24.00
Gross profit	40.00	32.00	70.00	56.00	40.00	32.00	70.00	56.00
– Selling, general & admin. (FC)	15.00	15.00	45.00	45.00	15.00	15.00	45.00	45.00
Operating profits	25.00	17.00	25.00	11.00	25.00	17.00	25.00	11.00
– Interest expense	0.00	0.00	0.00	0.00	15.00	15.00	15.00	15.00
Earnings before tax	25.00	17.00	25.00	1.00	10.00	2.00	10.00	–4.00
– Income taxes (35%)	8.75	5.95	8.75	3.85	3.50	0.70	3.50	–1.40
Net income	€16.25	€11.05	€16.25	€7.15	€6.50	€1.30	€6.50	–€2.60
Percentage change in net income		–32.0%		–56.0%		–80.0%		–140.0%

The exhibit shows how a decrease in revenue affects net income (total equity risk) for four different combinations of debt (financial risk) and operating leverage (business risk). In columns 1 and 2, we see the effect on a firm with no debt and low operating leverage; in columns 3 and 4, no debt and high operating leverage; in columns 5 and 6, debt and low operating leverage; and in columns 7 and 8, debt and high operating leverage. As you can see, total equity risk, represented by the percentage drop in net income, is greater when operating leverage is higher (for example, compare columns 1 and 2 with columns 3 and 4) and when a firm has financial risk (for example, compare columns 1 and 2 with columns 5 and 6). Furthermore, financial risk magnifies operating risk (for example, compare columns 3 and 4 with columns 7 and 8).

No Qs on this
just for visual
purposes

Exh 16.4: Illustrations of M&M Proposition 2

visual purposes

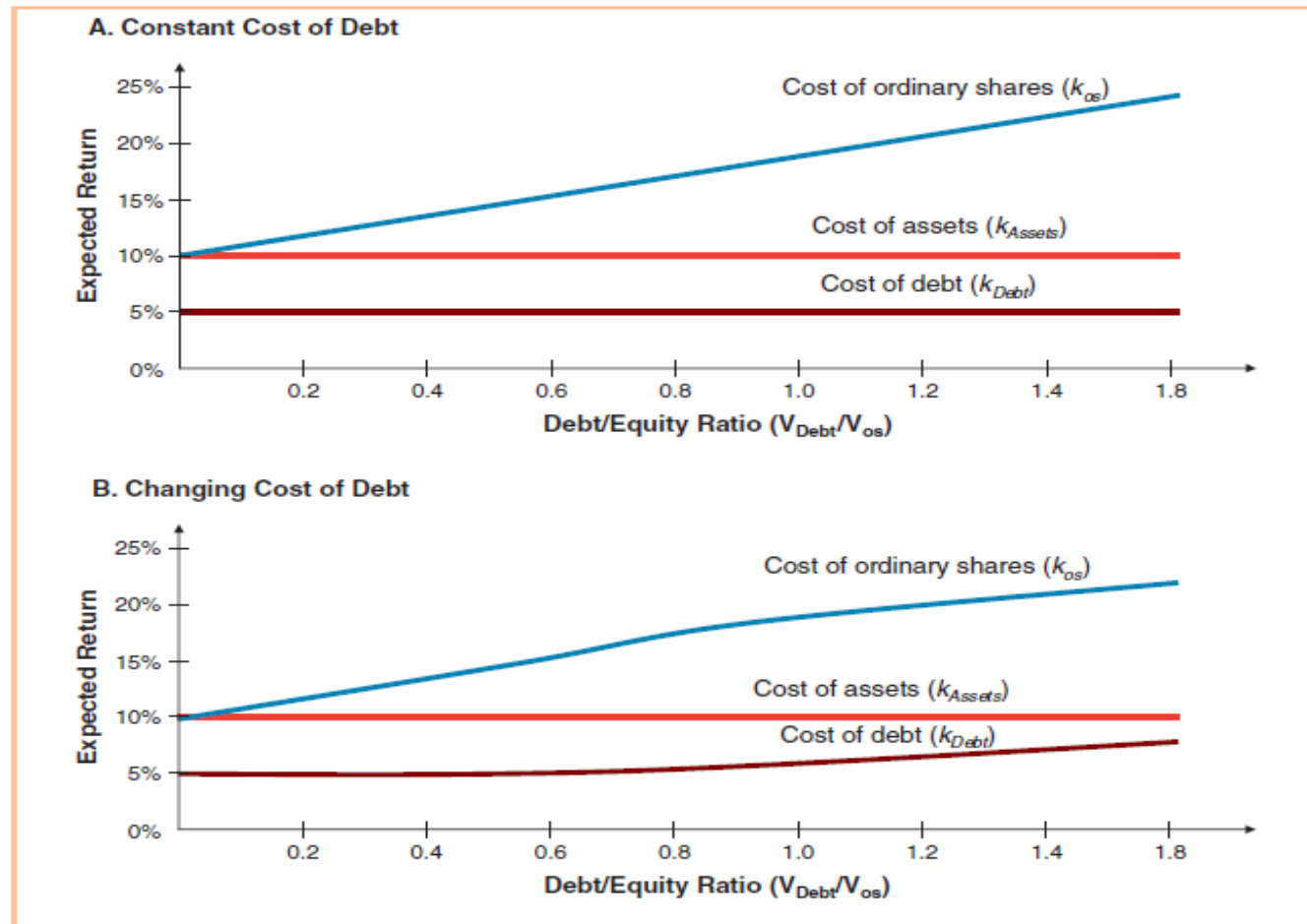


Exhibit 16.4: Illustration of M&M Proposition 2 The exhibit shows the cost (required return) of assets, ordinary shares and debt for different debt-to-equity ratios. Panel A assumes that the cost of debt remains constant and Panel B assumes that the cost of debt increases with leverage. The cost of assets, which is the return investors require to compensate them for the business risk, does not change with leverage. As M&M Proposition 2 tells us, the cost of ordinary shares increases with leverage.

The Modigliani and Miller Propositions

Proposition 2

If financial policy matters, it must be because

1. Taxes matter. *there is now gov*
2. Information or transaction costs matter.
3. Capital structure choices affect a firm's real investment policy.

The Modigliani and Miller with Taxes

- Modigliani and Miller adjusted their theory to recognize that the existence of income taxes did in fact mean that the value of a firm could be enhanced by having debt in its capital structure.
- In their initial conclusions (where there are no taxes), they proved that the Value of the levered Firm, i.e. $V_L = V_U$ (The Value of the unlevered Firm).
if there are no taxes
- That is, the value of a firm is the same whether it is levered or unlevered.
- Cost of Debt is tax deductible.
- The tax shield provided by debt creates value.

The Modigliani and Miller with Taxes

Under M-M, with company taxation, the value of the geared firm will be higher by present value of the interest tax shield.

$$VL = VU + \text{PV of expected interest tax shield}$$

Where: VL = Value of levered firm

VU = Value of unlevered firm

} know for test

The Modigliani and Miller with Taxes

Relaxing other assumptions:

- Too much debt increases the cost of financial distress.
- Too much debt could impact the structure of the balance sheet and reduce EBIT.
- Too much debt will incur agency costs.
- Too much debt will lead to an increase in K_d .

The Modigliani and Miller with Taxes

Taking these factors into account, a more accurate equation can be derived:

$$VL = VU + PVI - PVF - PV\Delta EBIT - PVA - PVD$$

PVI = PV of expected interest tax shield

PVF = PV of expected financial distress

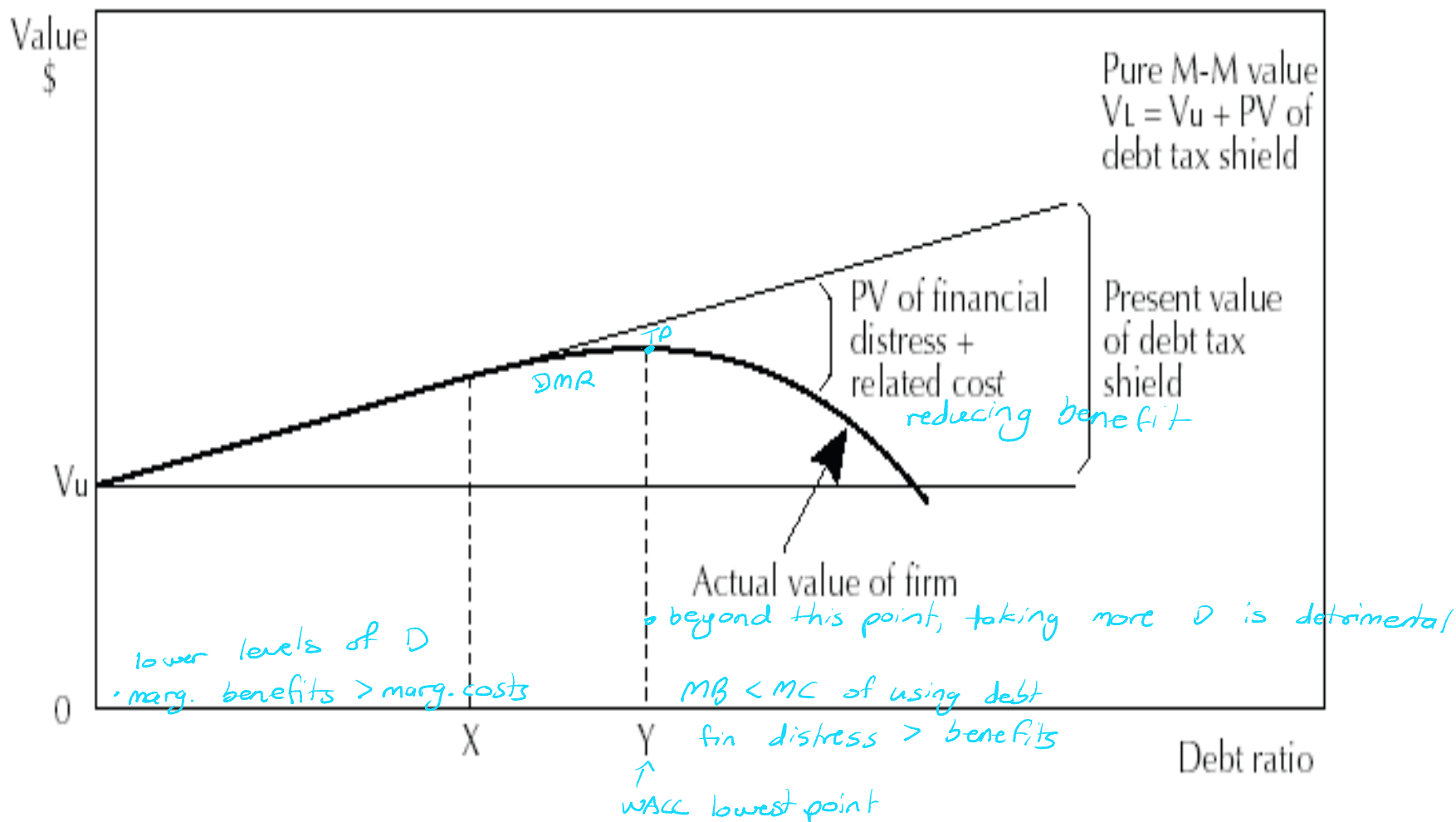
$PV\Delta EBIT$ = Reduction in value from lower EBIT

PVA = Reduction in value from agency costs

PVD = Reduction in value from increased cost of debt

The Modigliani and Miller with Taxes

The relationship in the previous slide is reflected in the figure below



The Modigliani and Miller with Taxes

From the previous figure, you can see that the interest tax shield gives rise to an increase in value but that from point X, the increase in value declines as expected costs of financial distress and related costs come into play.

Point Y is the point where the marginal benefit from any interest tax shield is exactly offset by the increase in the marginal cost. This point is the optimal capital structure.

Thus, any further increases in debt beyond Y will result in the marginal disadvantages being bigger than the marginal tax shield.

Note, that while it is possible to estimate the present value of the tax shield fairly accurately, the other factors are difficult to estimate. For example, the cost of financial distress is difficult to estimate because it depends on numerous factors

Note that the formula on slide 36 does not give us a model that can be used in practice to determine the optimal capital structure but it provides us with a comprehensive conceptual analysis which has given rise to the trade off theory.

Exh 16.5: Capital Structure and Firm Value with Taxes

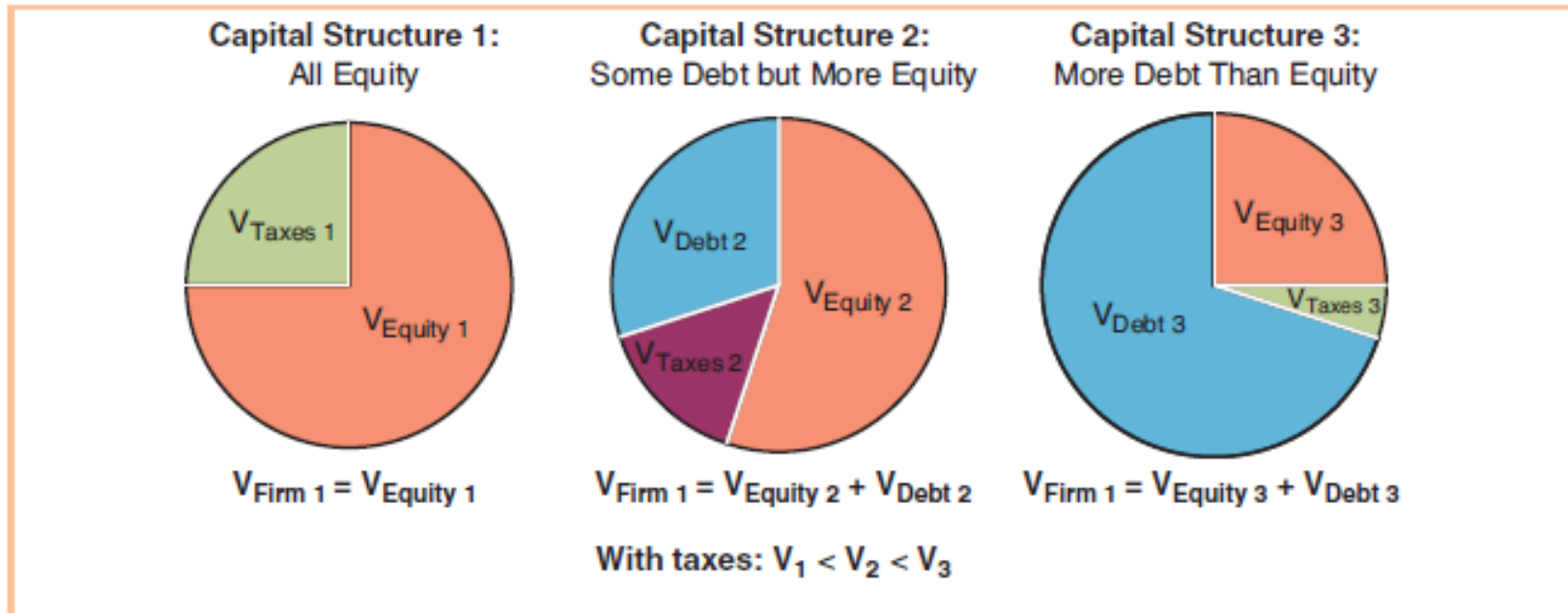


Exhibit 16.5: Capital Structure and Firm Value with Taxes Leverage can increase the value of the firm when interest payments are tax deductible but dividend payments are not. The pie on the left represents a firm financed 100% via equity. The slice labelled $V_{\text{Taxes } 1}$ reflects the proportion of the cash flows from operations that this firm pays in taxes. The two pies to the right illustrate how the value of the cash flows paid in taxes decreases as leverage is increased. By reducing the fraction paid in taxes, leverage increases the value of the firm in these examples.

The Benefits and Costs of Using Debt

- The total amount of interest paid each year, and therefore the amount that will be deducted from the firm's taxable income, is

$$D \times k_{\text{Debt}} = \text{interest amt}$$

cost of debt

- This will result in a reduction in taxes paid (the interest tax shield) of $D \times k_{\text{Debt}} \times t$, where t is the firm's marginal tax rate that applies to the interest expense deduction.

The Benefits and Costs of Using Debt

- If this reduction will continue in perpetuity, the perpetuity model, to calculate the present value of the tax savings from debt, is:

$$V_{\text{Tax savings debt}} = PVA_{\infty}$$

$$= \frac{CF}{i}$$

$$= \frac{D \times k_{\text{Debt}} \times t}{i} \quad \text{tax shield}$$

i AT → convert to i BT

$$B4 \quad k_{\text{debt}}(i) = \frac{k_{\text{debt AT}}}{(1-t)}$$

The Benefits and Costs of Using Debt

- Since the firm will benefit from the interest tax shield only if it is able to make the required interest payments, the cash savings associated with the tax shield are about as risky as the cash flow stream associated with the interest payments.

$$V_{\text{Tax-savings debt}} = D \times T \quad (16.4)$$

$$\begin{aligned} V_{\text{Tax savings debt}} &= PVA_{\infty} \\ &= \frac{CF}{i} \\ &= \frac{D \times k_{\text{Debt}} \times t}{i} \end{aligned}$$

Debt and Tax shield Example 1

- Assume that Padua Ltd invests R1000 million in operating assets and management finances this with a loan. Assume that tax rate is 28% and the cost of debt is 10%.
- If the firm pays interest of R100 million, then this will result in a tax deduction of R100 million and a tax shield of R28 million per year. If the financing stays in place or is indefinite, then we will receive a tax shield each year in the future.
- Calculate the value of the tax shield and the value of the firm

$$\frac{D \times IR \times t}{i}$$

$$\text{or } D \times t = 280$$

$$PV \text{ tax shield} = 280$$

Debt and Tax shield Example 1 solution

- Using the formula in slide 39

$$\begin{aligned} V_{\text{Tax savings debt}} &= PVA_{\infty} \\ &= \frac{CF}{i} \\ &= \frac{D \times k_{\text{Debt}} \times t}{i} \end{aligned}$$

- Value of tax shield = $D \times t$
- $CF = R28$ million and i is 10%
- Value of the tax shield = $R28 \text{ million} / 0,1$
 $= 280$ million

Or

$$\begin{aligned} \text{Value of the tax shield} &= [1000m \times 0,1 \times 0,28] / 0,1 \\ &= R280 \text{ million} \end{aligned}$$

Or

$$\text{Value of the tax shield} = R1000m \times 0,28 = R280m$$

Debt and Tax shield Example solution 1 continued

- Now going back to M-M, If M-M applies except for taxation, then we can obtain the value of the firms as follows
- Value of the firm $= V_L$ = value of firm financed with all equity V_U (i.e. value of the unlevered firm) + value of the tax shield $PVTS$
- In this example market value of the all equity V_e would be equal to the value of the assets, that is R1000 million.
- Value of the tax shield is equal to R280 million, hence, using the calculation of the valuation of the levered firm above, the value of the firm (levered) is
- $= R1\ 000m + R280m = R1\ 280m$
- Note that in this example we assumed a permanent level of debt financing, we used the growing perpetuity formula to determine the value of tax shields

** Test, just numbers change* Debt and Tax shield Example 2 *1:46:30*

- Assume that ABC Ltd has assets with a market value of R1000million which are currently financed 100% with ordinary equity. Assume that the company is selling the business as a going concern to a Newco and ABC Ltd owns 100% of the shares of the Newco, which means the operating company will be placed within Newco.
- Assume that the bank will lend R900 million to Newco resulting in a highly levered company in order to maximize the interest tax shields. Assume that ABC Ltd will provide some security to the bank to ensure that interest rates are set at a reasonable level. Assume interest rate is 12%.
- Assume also that the tax consequences of the transfer of the business assets to Newco are insignificant and focus only on the tax shields arising from the levered structure.
- Given the above, Newco will receive the loan from the bank R900 million and pay this over to ABC Ltd in consideration of the assets acquired. Assume also that ABC Ltd will invest equity financing of R100 million in Newco.
- The financial statements of the operating company before and after the transaction will look as follows:

Debt and Tax shield Example 2 solution

	ABC Ltd	Newco
Balance Sheet		
None Current Assets	700	700
Other Assets		
Net Working Capital	300	300
Total Assets	1000	1000
Equity	1000	100
Long tem loans	0	900
	1000	1000
Income statement		
EBIT	200	200
Less: Interest expense	0	-108
	200	92
Less: Taxation @28%	-56	-25,8
Net Income	144	66,2
ROA $\text{NOPAT} = \text{EBIT} (1-T)$	14,4%	14,4%
ROE	14,4%	66,2%

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Debt and Tax shield Example 2 solution

From the previous slide

The ROA of **14,4%** (i.e. $(200(1-0,28)/1\ 000)$) did not Change but the ROE changed significantly from **14.4%** (i.e. $144/1000$) to **66.2%** ($66,2/100$).

If M-M applies except for taxation, we can determine the value of the firm as follows:

Value of the company = value of the firm if all equity financed + value of tax shield

In our example the value of the tax shield = **R252 million**

Calculated as $D \cdot t$ or $[D \cdot K_d \cdot t]/K_d$ where $D \cdot K_d \cdot t$ is the tax shield and K_d is the cost of debt, in this case **12%**

Thus, $900 \cdot 0,12 \cdot 0,28 = \text{tax shield} = \text{R30,24 million}$, and this benefit is in perpetuity, thus, the present value of the tax shield is valued as a perpetuity

PV tax shield = $30,24/0,12 = \text{R252million}$,

which is the same as $D \cdot t = 900\text{million} \cdot 0,28 = \text{R252million}$ D E PVS

Thus the total value in the group (levered firm) = $R900 + R100 + R252$
= R1 252million